INCH-POUND

MIL-PRF-5501/31A
15 January 1997
SUPERSEDING
MIL-C-5501/31
29 July 1994

PERFORMANCE SPECIFICATION SHEET

CAP, CIRCULAR, ESD, PROTECTIVE, ELECTRICAL CONNECTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-C-5501 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation

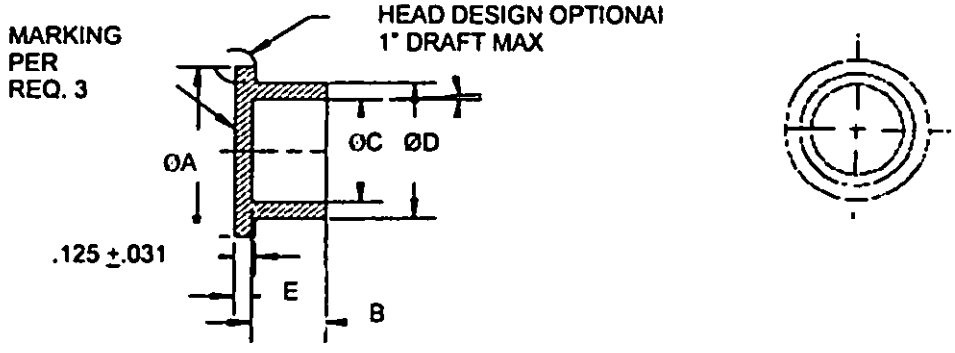


TABLE 1. Circular caps and dimension.

DASH NO.	ØA ±.031	B ±.031	ØC ±.010	ØD ±.010	E ±.010
250	.670	.500	.250	.336	.040
313	.670	.500	.313	.399	.040
375	.670	.500	.375	.461	.040
413	.670	.625	.413	.483	.040
480	.740	.500	.480	.540	.040
500	.750	.500	.500	.586	.040
530	.750	.625	.530	.595	.040
605	.840	.500	.605	.665	.040
655	.860	.625	.655	.715	.040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE 1. Circular caps and dimension - (Cont'd).

DASH NO.	ØA ±.031	B ±.031	ØC ±.010	ØD ±.010	E ±.010
720	.970	.500	.720	.775	.040
785	1.031	.625	.785	.845	.045
850	1.125	.500	.850	.905	.045
893	1.125	.625	.893	.963	.045
973	1.220	.500	.973	1.035	.045
1028	1.250	.625	1.028	1.088	.045
1093	1.330	.562	1.093	1.155	.045
1125	1.400	.562	1.125	1.213	.045
1150	1.406	.625	1.150	1.215	.050
1231	1.465	.562	1.231	1.301	.050
1270	1.500	.625	1.270	1.340	.050
1340	1.590	.562	1.340	1.410	.050
1390	1.640	.625	1.390	1.460	.050
1463	1.700	.562	1.463	1.533	.050
1540	1.810	.625	1.540	1.580	.050
1645	1.875	.625	1.645	1.718	.055
1715	1.950	.562	1.715	1.790	.055
1890	2.125	.625	1.890	1.970	.055
1965	2.220	.562	1.965	2.040	.055
2140	2.340	.625	2.140	2.210	.055
2215	2.435	.600	2.215	2.290	.055
2380	2.600	.625	2.380	2.450	.060
2440	2.660	.600	2.440	2.530	.060
2630	2.875	.625	2.630	2.700	.060
2720	2.960	.600	2.720	2.812	.060
2880	3.125	.625	2.880	2.950	.060
2960	3.225	.600	2.960	3.050	.060

REQUIREMENTS:

1. MATERIAL - The cap shall be made of a permanent, electrically conductive or static dissipative, non-sloughing material. Topical antistats shall not be used. The material shall have a volume resistance of less than or equal to 10^9 ohms measured as specified in note 2.
2. RESISTANCE TO SLOUGHING/CRAYONING - When rubbed briskly against a hard, flat laminated surface, the caps shall exhibit no tendency to slough or mark the surface in a manner similar to a crayon.

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3. MARKING - Manufacturer name, EIA RS-471 electrostatic discharge symbol (hand in triangle) per MIL-STD-1686, and part number shall be marked on the outer face of the caps. Caps having an inner diameter (Dimension "ØC") of less than .500 inches do not require the symbol.
4. PART NUMBER - The part number consists of M5501/31A and a dash number taken from table 1 (example: M5501/31A - 250).

NOTES:

1. Unless otherwise specified, dimensions are in inches.
2. Volume resistance shall be measured as follows:
    a. The test apparatus shall consist of an electrode, a calibrated ohmmeter and a non-oxidizing metal sheet having adequate surface area to accommodate the entire tested cap circumference. The electrode shall be constructed in accordance with ASTM F150 with the following dimensional exceptions: the electrode and rubber insert diameter shall be 1.5 ±0.05 inches and the total weight shall be 2.0 lb ±0.1 oz. The ohmmeter shall be capable of supplying an open circuit test voltage of between 5 and 15 volts and shall be capable of measuring resistances of between 0 and 1,000,000,000 ohms. The non-oxidizing metal shall be flat, free of any insulative coatings and at least 1/16 inch thick.
    b. All test samples shall be preconditioned at 10 ±2% RH and 70 ±2°F for 24 hours prior to testing in the same environment.
    c. Resistance measurements shall be made by laying the test cap on the metal sheet (open side up) and placing the electrode inside the cap with the aluminum foil side contacting the cap. Each measurement shall be made by connecting the leads of the ohmmeter to the top of the electrode and the metal plate and applying the test voltage. The resistance of the cap shall be as specified.
3. Sampling/Testing - One cap shall be selected at random per 1000 of each size purchased. When less than 1000 caps are purchased, one sample shall be selected at random for testing. For purchased caps having a diameter (Dimension "ØC") smaller than the test electrode, a cap from the same lot having a diameter greater than 1.5 inches shall be used.
4. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.
5. Referenced Government (or non-Government) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.
6. Electrically conductive or static dissipative material is defined as having a volume resistance of less than or equal to 10^9 ohms.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - SH
Air Force - 11

Review activities:
Air Force - 70, 71, 80, 82, 84, 99
DLA - IS

Preparing activity:
Air Force - 11

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